RLVR · DEMO · UNIFIED REPORT

RLVR on small instruct models — four runs

Companion to the SC-AI Seminar talk on RLVR (April 2026) · Tommie & Lan Li

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RLVR DEMO · FOUR RUNS · ONE NARRATIVE

Four RLVR runs. One that learns.

A walk through what had to go right.

Bottom line: RLVR training is a tough cookie. Modern instruct models are already strong — adding on top requires **three** variables to line up at once: baseline in the variance band, enough non-dead groups, and a policy update budget that actually lets the policy move. Runs 1–3 each missed one. Run 4 hit all three — and the KL ceiling is near the drift zone even so.

Headline

RUN	TASK	BASE MODEL	G	BASELINE PASS@1	POST-RLVR PASS@1	Δ	KL MAX
Run 1 · Code (OLMo)	MBPP	`OLMo-2-1124-7B-Instruct`	4	2.72%	3.11%	+0.39pp	0.0009
Run 2 · Math (OLMo)	GSM8K	`OLMo-2-1124-7B-Instruct`	4	82.64%	82.11%	-0.53pp	0.0021
Run 3 · Math (Gemma)	GSM8K	`gemma-2-2b-it`	8	58.53%	58.23%	-0.30pp	0.0011
Run 4 · Math (Gemma v2)	**GSM8K**	**`gemma-2-2b-it`**	**8**	**58.53%**	**60.35%**	**+1.82pp**	**0.0323**

The three constraints GRPO needs

1. **Baseline in the variance sweet spot.** Not so low that every completion fails (Run 1: 2.7%), not so high that every completion passes (Run 2: 82.6%).

2. **Enough non-dead groups.** For baseline p and group size G, the all-same share is $p^G + (1-p)^G$. Empirically per-prompt difficulty clusters outcomes, so the observed share is higher than the independence model predicts.

3. **A policy update budget that lets the policy move.** Learning rate, KL coefficient β, step count. Runs 1–3 all used lr=5e-6, β=0.05, 200 steps — and the policy never meaningfully moved (Run 3's KL max was 0.0011). Reward variance was necessary but not sufficient.

Per-run diagnosis

RUN 1 · CODE (OLMO)

Baseline too low (2.7%) — sparse reward, no gradient.

Read the full run →

RUN 2 · MATH (OLMO)

Baseline too high (82.6%) — groups saturate all-pass.

Read the full run →

RUN 3 · MATH (GEMMA)

Variance fixed; update budget still capped. KL max 0.0011.

Read the full run →

RUN 4 · MATH (GEMMA V2)

Same recipe as Run 3 with lr×4, β÷10, steps×2. +1.8pp — first signal outside noise.

Read the full run →

The takeaway

Run 4 validated the hypothesis. Same model, same G, same reward setup as Run 3 — only the policy update budget changed (lr 4×, β ÷10, steps 2×). KL max went from 0.0011 → **0.032** (29×). pass@1 went **58.53% → 60.35% (+1.82pp)**, the first delta outside noise across all four runs. **120 fixed, 96 regressed**, net +24.

But KL max 0.032 is high enough that **drift is real** — 96 problems the base model used to solve are now broken. On a single-task benchmark that's still a net win; on a multi-task eval it would be the place to watch for collateral damage.

Honest caveat

> Run 4's reward curve is noisy and drifts *slightly* down in the tail. Per-step reward is averaged over ~16 completions (2 prompts × G=8), so noise dominates the trend. The pass@1 number on 1,319 problems is the smoothed signal — that's where the +1.82pp lives. Worth flagging.

Read the runs

- Run 4 — Gemma/GSM8K, update budget unlocked — the only run that clearly learns.
- Run 3 — Gemma/GSM8K, variance band — the setup that Run 4 builds on.
- Run 2 — OLMo/GSM8K — baseline too high.
- Run 1 — OLMo/MBPP — the original talk demo; baseline too low.

RUN 1 · MBPP · OLMO-2-7B-INSTRUCT

Code: baseline too low

Baseline too low (2.7%) — sparse reward, no gradient.

MBPP is a code-completion benchmark where OLMo-2-7B-Instruct (not code-tuned) solves only 2.7% of held-out problems. With G=4, the fraction of groups where every completion fails is (1−0.027)^4 ≈ 89%. Most optimizer steps carry no signal. Δ = +0.4pp is within noise.

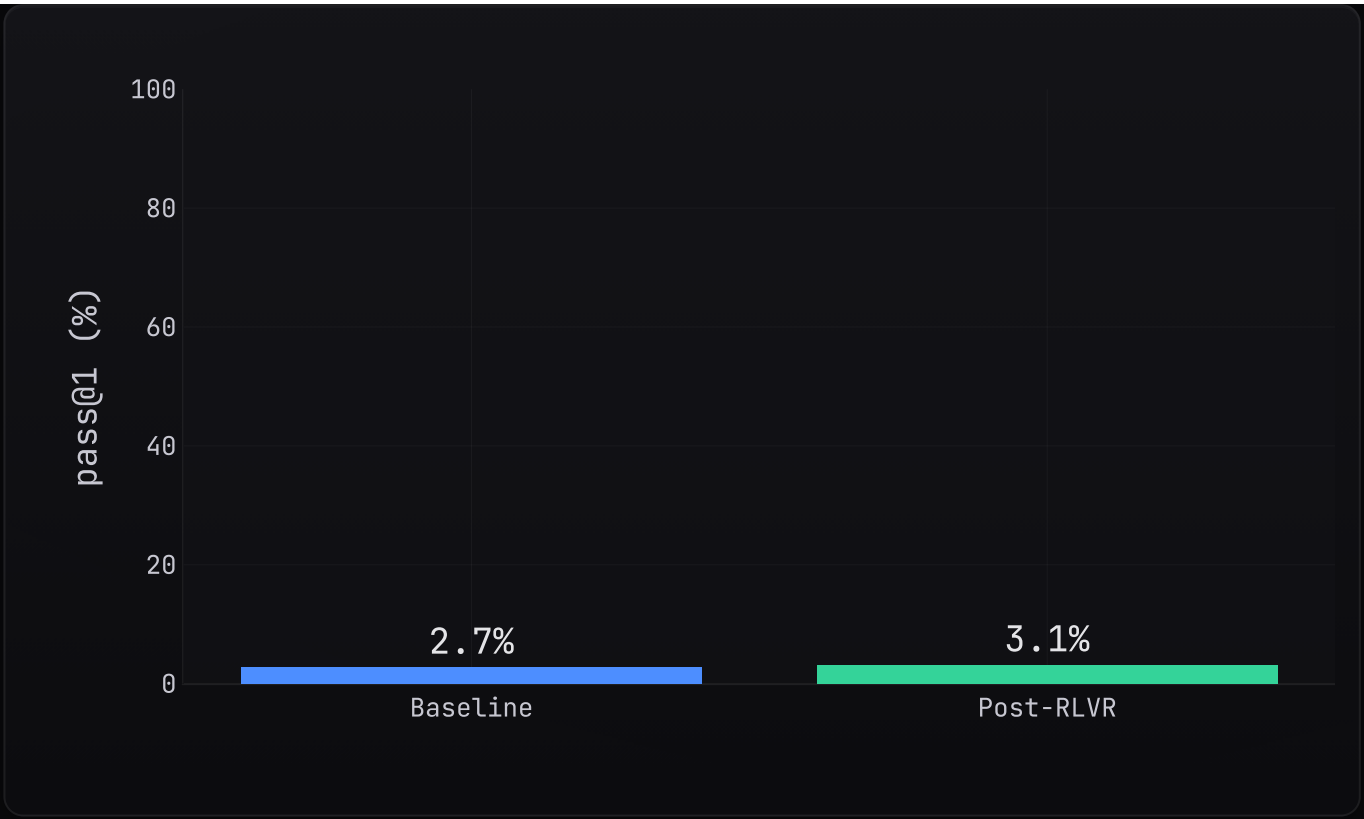
2.72%	3.11%	+0.39pp
BASELINE PASS@1	POST-RLVR PASS@1	Δ

200	0.0009
TRAINING STEPS	KL MAX

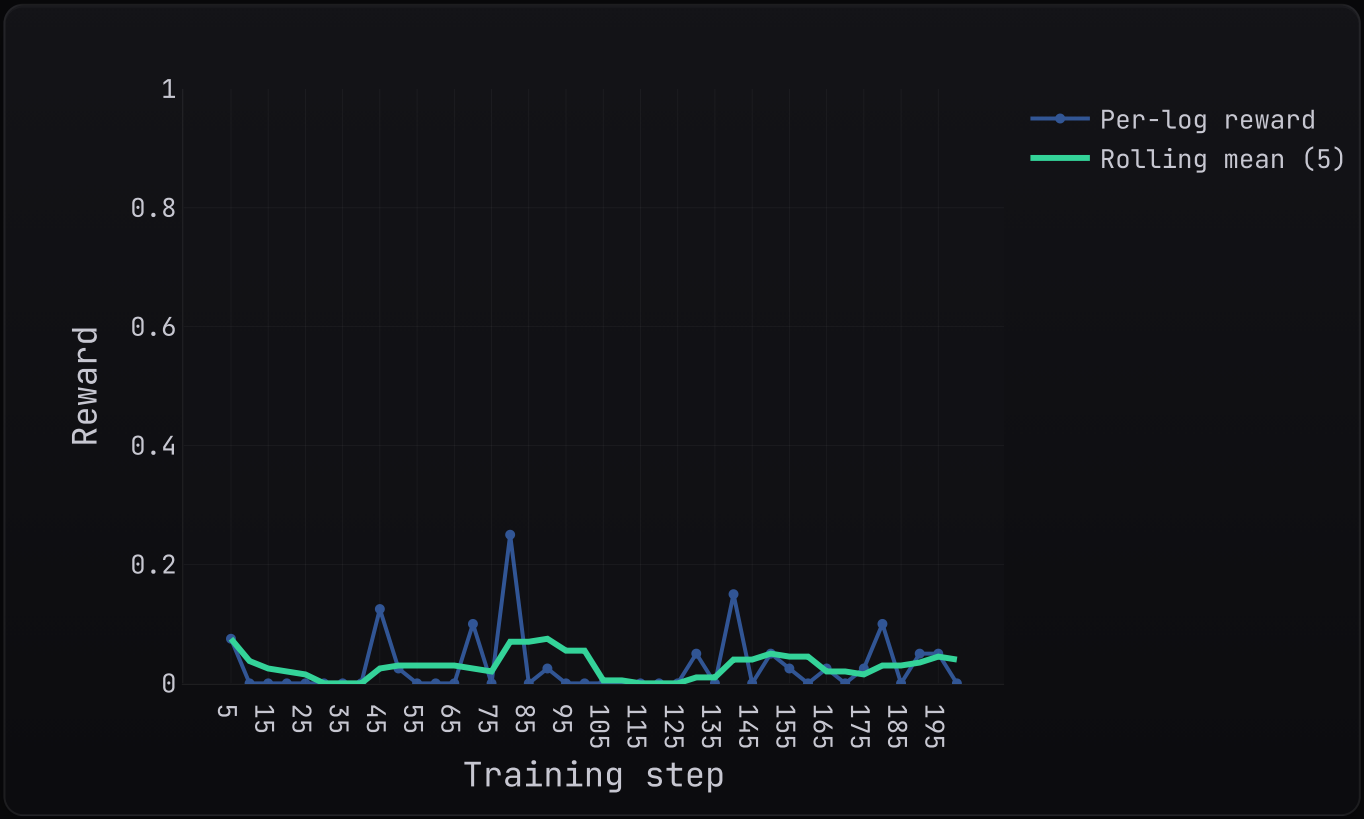
Recipe

BASE MODEL	allenai/OLMo-2-1124-7B-Instruct
GENERATIONS PER PROMPT (G)	4
TEMPERATURE	1.0
TRAINING STEPS	200
LEARNING RATE	5e-06
KL COEFFICIENT B	0.05
LORA R / A	16 / 32
MAX COMPLETION TOKENS	512
VARIANCE BAND (PREDICTED)	10.5% mixed-reward

pass@1 on MBPP held-out (257)



Reward (fraction of completions with correct answer)



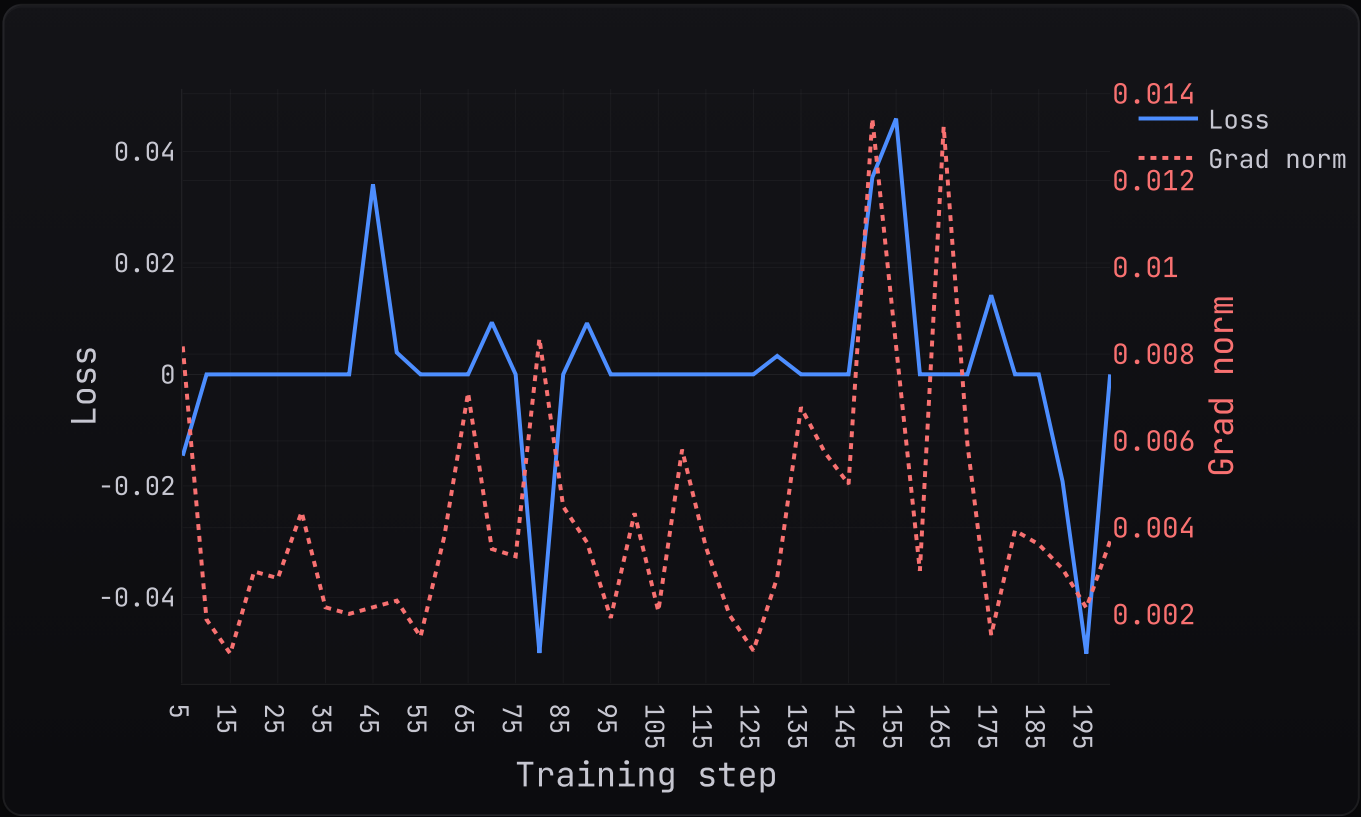
Reward-variance diagnostic · `frac_reward_zero_std`

Percent of GRPO groups where all G completions got the same reward. These groups have zero advantage and contribute no gradient. Run average: **95.8%**.

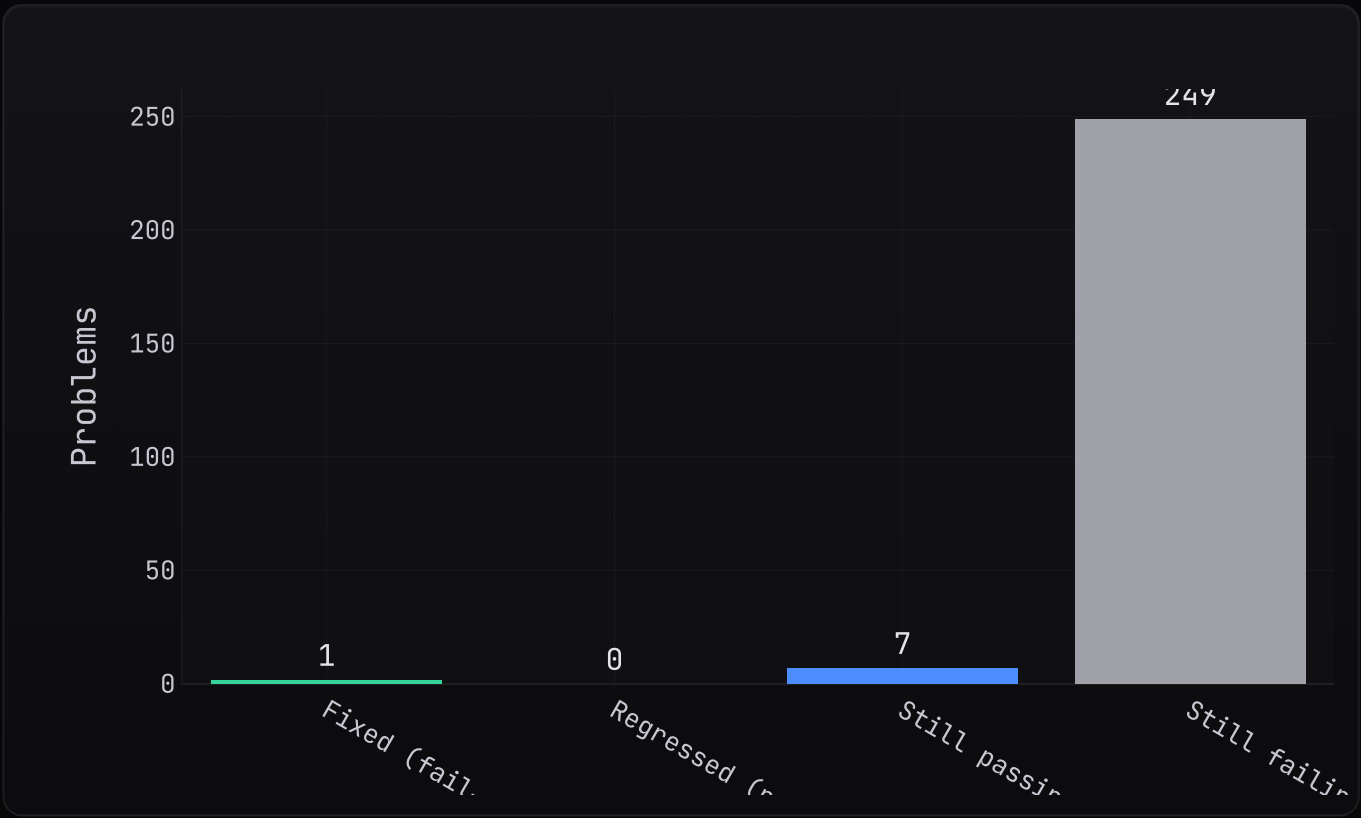
KL divergence from reference policy

Low = trained policy stays close to base. High = drift — watch for degradation.

Policy loss and gradient norm



Per-problem flip analysis (257 common problems)



RUN 2 · GSM8K · OLMO-2-7B-INSTRUCT

Math on OLMo: baseline too high

Baseline too high (82.6%) — groups saturate all-pass.

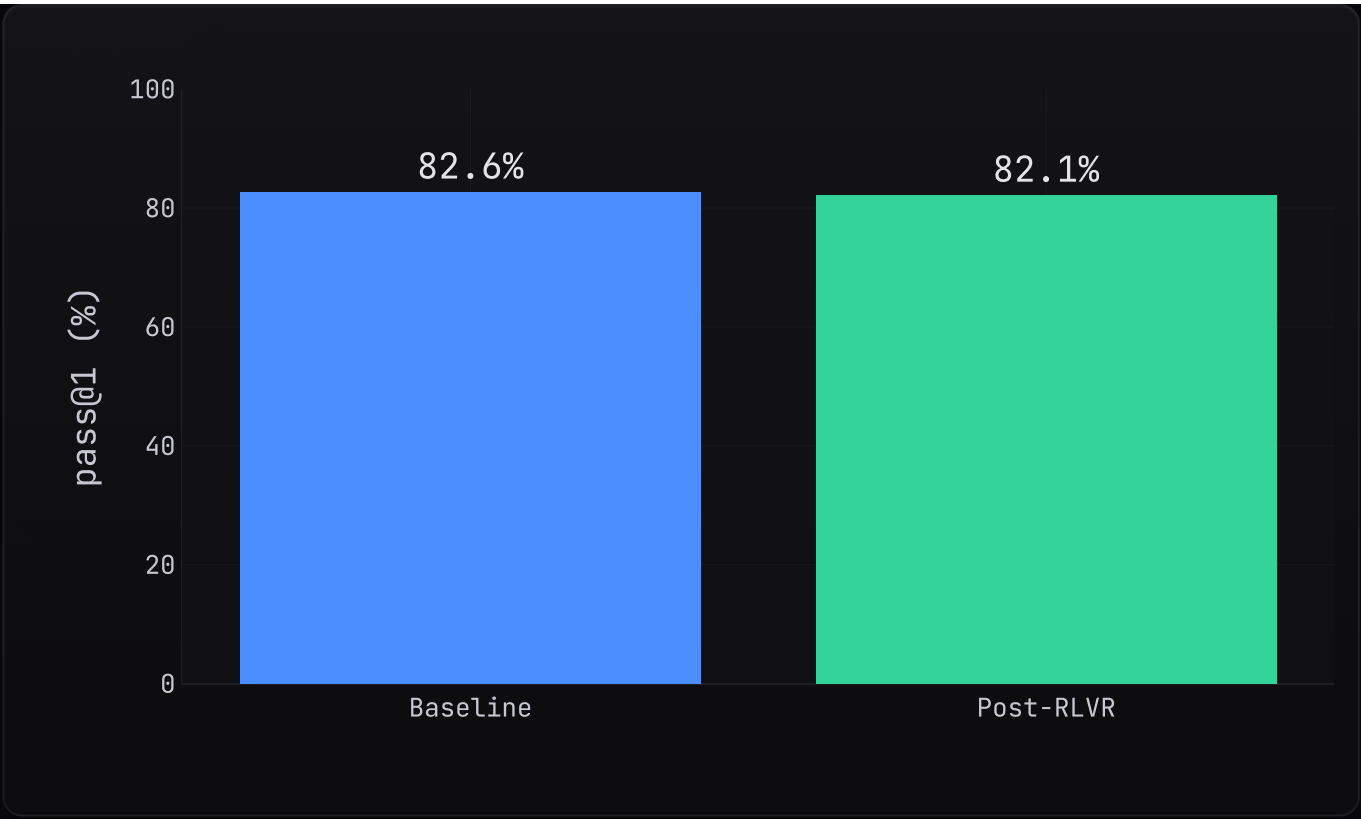
Same base model, same recipe, different task. GSM8K math is a sweet spot for OLMo-2's Tulu 3 fine-tune, so ~83% of completions pass — meaning most groups are all-pass, reward variance collapses, and the advantage term is zero most of the time. Δ = −0.5pp is within noise.

82.64%	82.11%	-0.53pp
BASELINE PASS@1	POST-RLVR PASS@1	Δ
200	0.0021	
TRAINING STEPS	KL MAX	

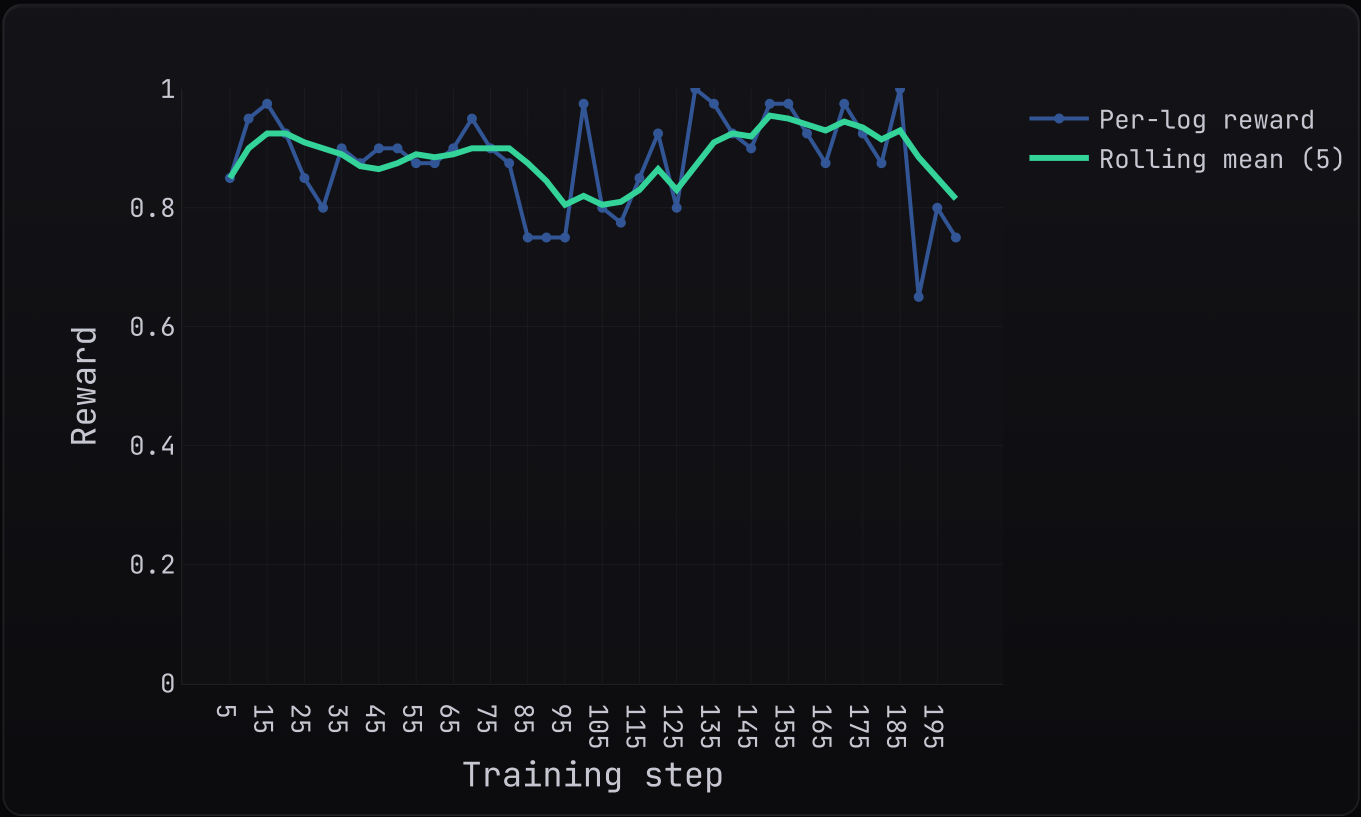
Recipe

BASE MODEL	allenai/OLMo-2-1124-7B-Instruct
GENERATIONS PER PROMPT (G)	4
TEMPERATURE	?
TRAINING STEPS	200
LEARNING RATE	5e-06
KL COEFFICIENT B	0.05
LORA R / A	16 / 32
MAX COMPLETION TOKENS	512
VARIANCE BAND (PREDICTED)	53.3% mixed-reward

pass@1 on openai/gsm8k test (1,319)



Reward (fraction of completions with correct answer)



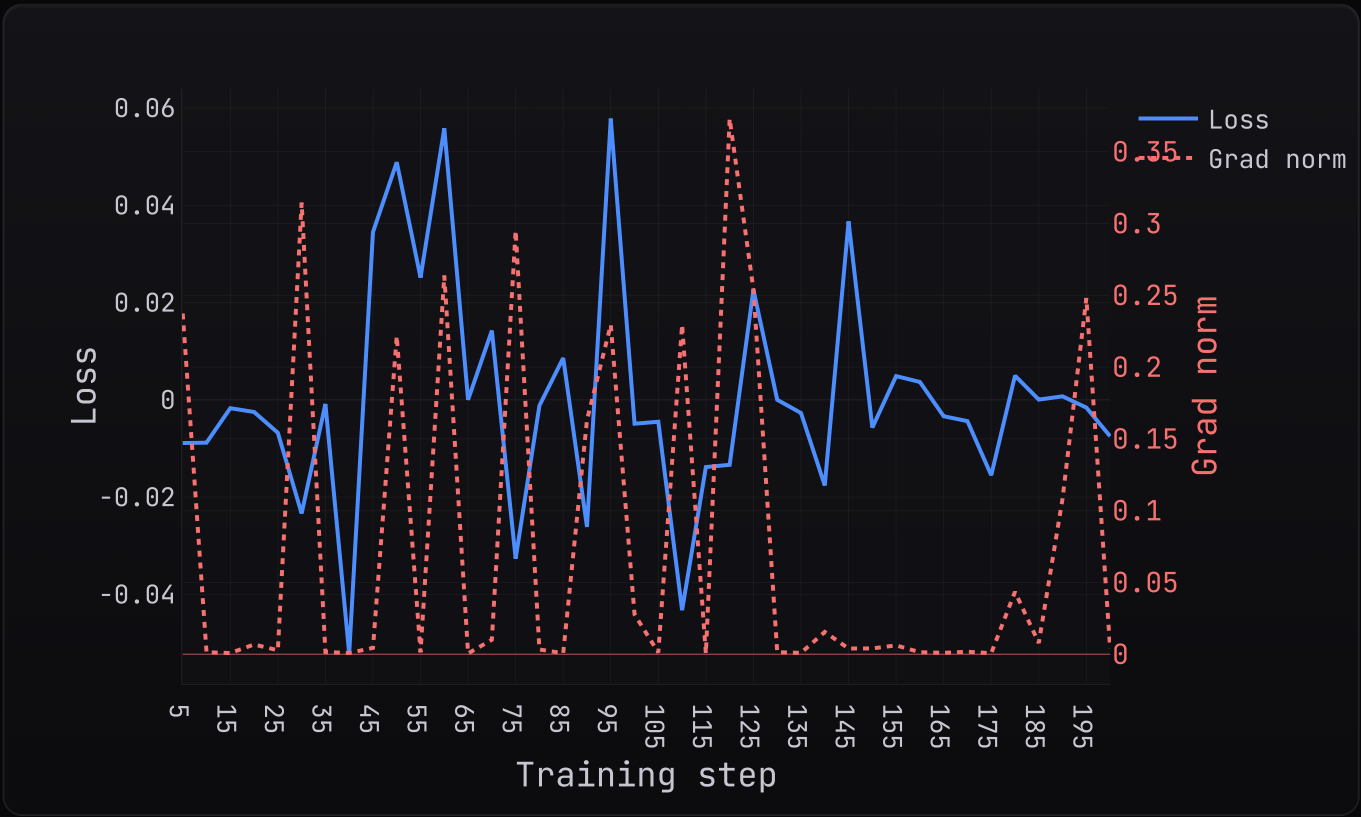
Reward-variance diagnostic · `frac_reward_zero_std`

Percent of GRPO groups where all G completions got the same reward. These groups have zero advantage and contribute no gradient. Run average: **80.5%**.

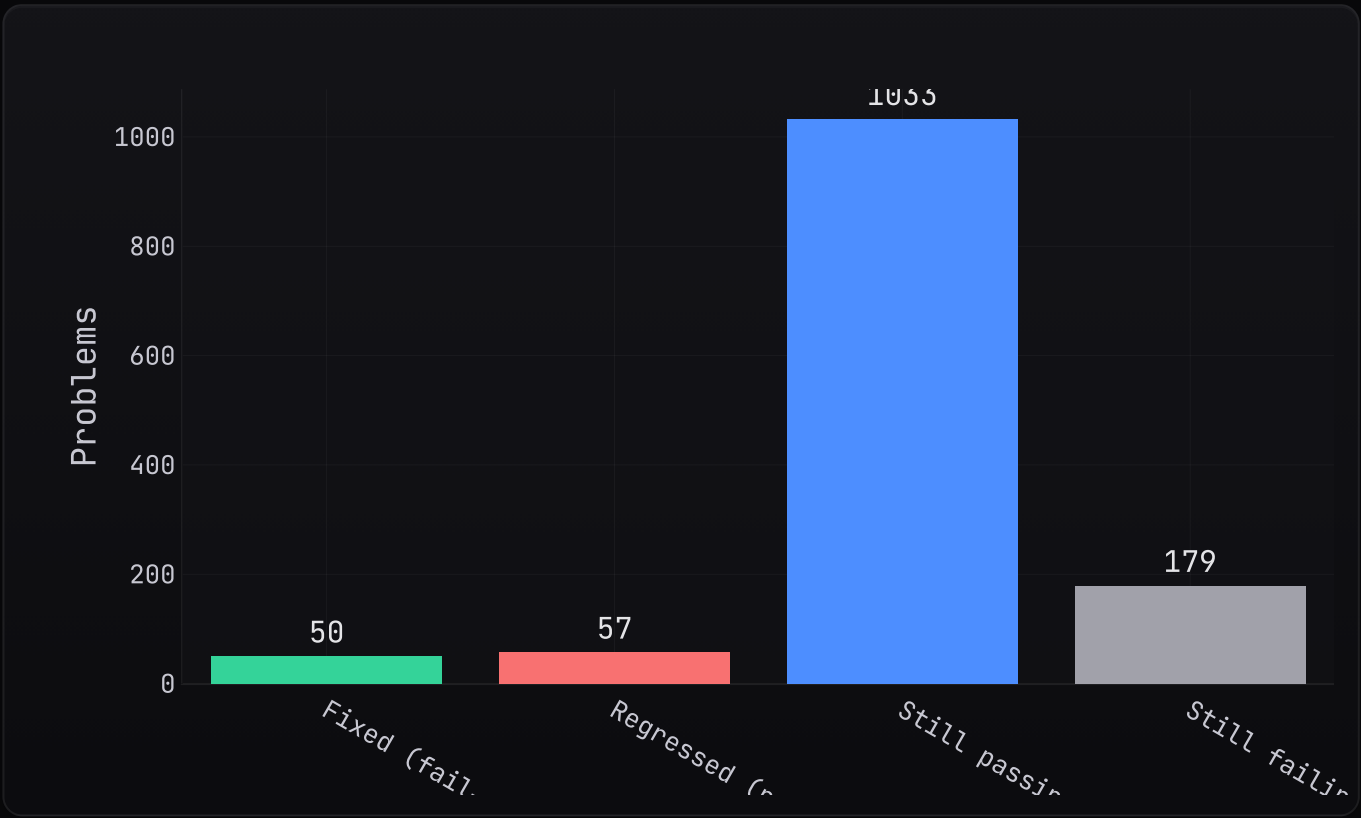
KL divergence from reference policy

Low = trained policy stays close to base. High = drift — watch for degradation.

Policy loss and gradient norm



Per-problem flip analysis (1319 common problems)



RUN 3 · GSM8K · GEMMA-2-2B-IT · G=8

Variance fixed · update budget still capped

Variance fixed; update budget still capped. KL max 0.0011.

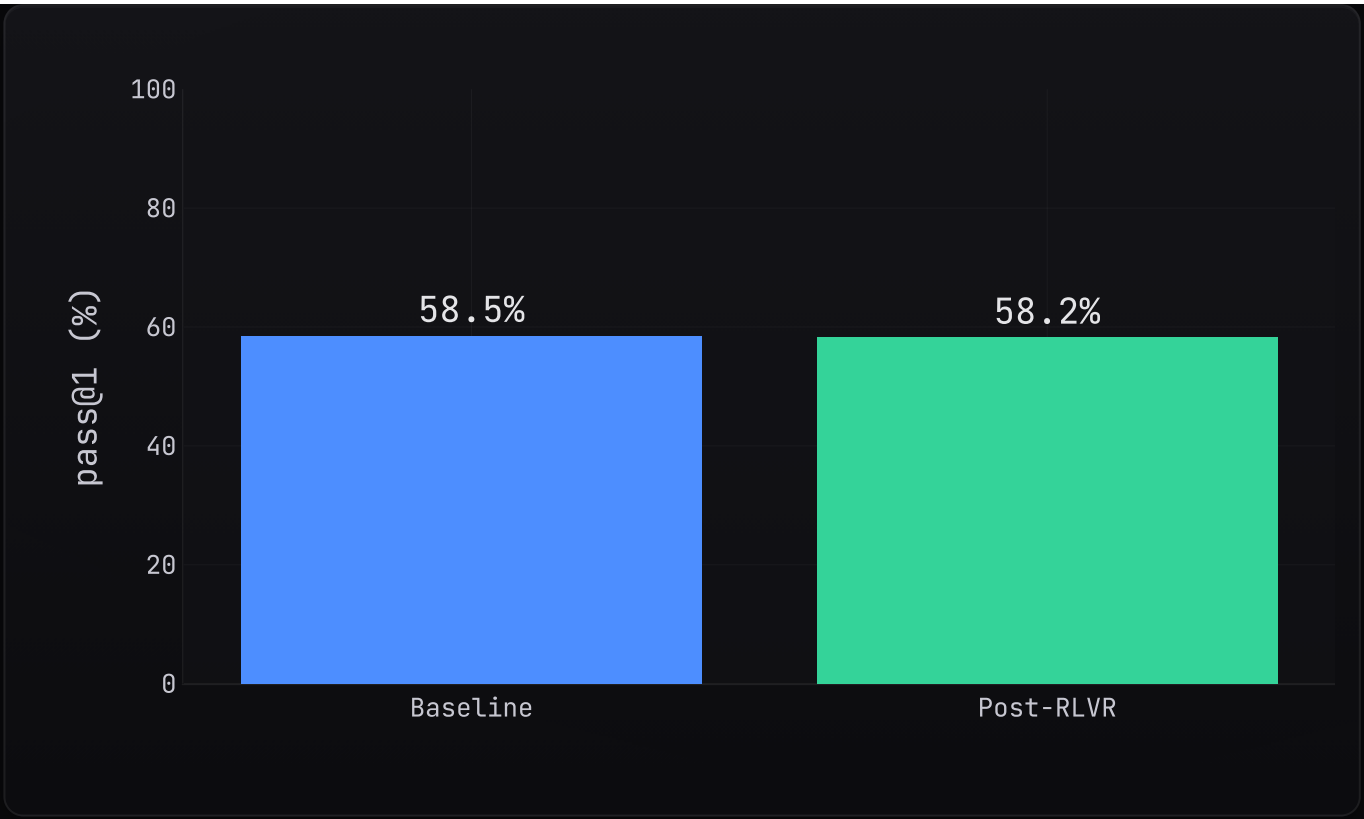
Swapped to Gemma-2-2B-IT (GSM8K baseline 58.5%, smack in the middle), bumped G=8 and temperature 1.0. Observed zero-variance share fell from ~80% (OLMo) to ~52% — about 2.4× more signal per step. But with lr=5e-6, β=0.05, 200 steps, the policy barely moved: KL max 0.0011. Δ = −0.3pp, still within noise.

58.53% BASELINE PASS@1	58.23% POST-RLVR PASS@1	-0.30pp Δ
200 TRAINING STEPS	0.0011 KL MAX	

Recipe

BASE MODEL	google/gemma-2-2b-it
GENERATIONS PER PROMPT (G)	8
TEMPERATURE	1.0
TRAINING STEPS	200
LEARNING RATE	5e-06
KL COEFFICIENT B	0.05
LORA R / A	16 / 32
MAX COMPLETION TOKENS	512
VARIANCE BAND (PREDICTED)	98.5% mixed-reward

pass@1 on openai/gsm8k test (1,319)



Reward (fraction of completions with correct answer)

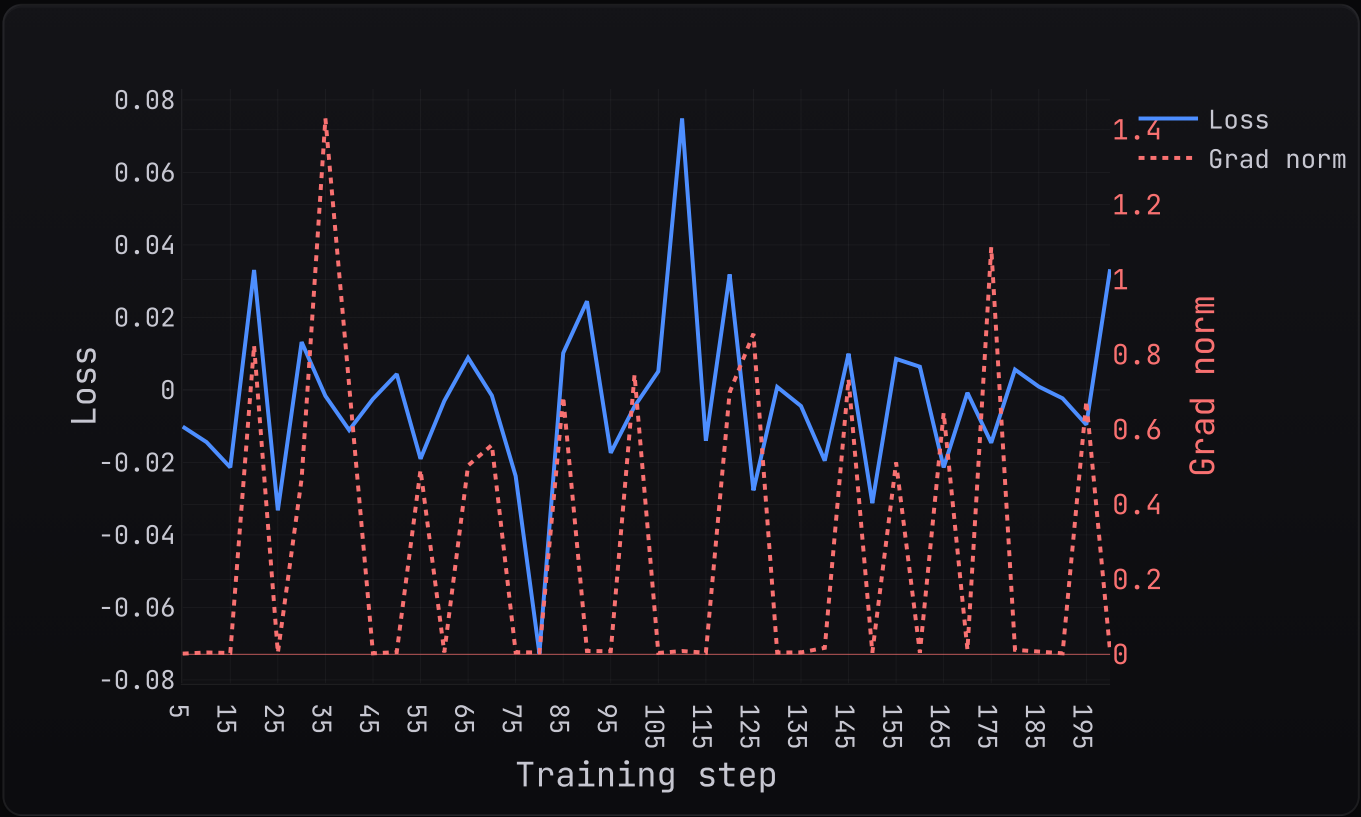
Reward-variance diagnostic · `frac_reward_zero_std`

Percent of GRPO groups where all G completions got the same reward. These groups have zero advantage and contribute no gradient. Run average: **52.5%**.

KL divergence from reference policy

Low = trained policy stays close to base. High = drift — watch for degradation.

Policy loss and gradient norm



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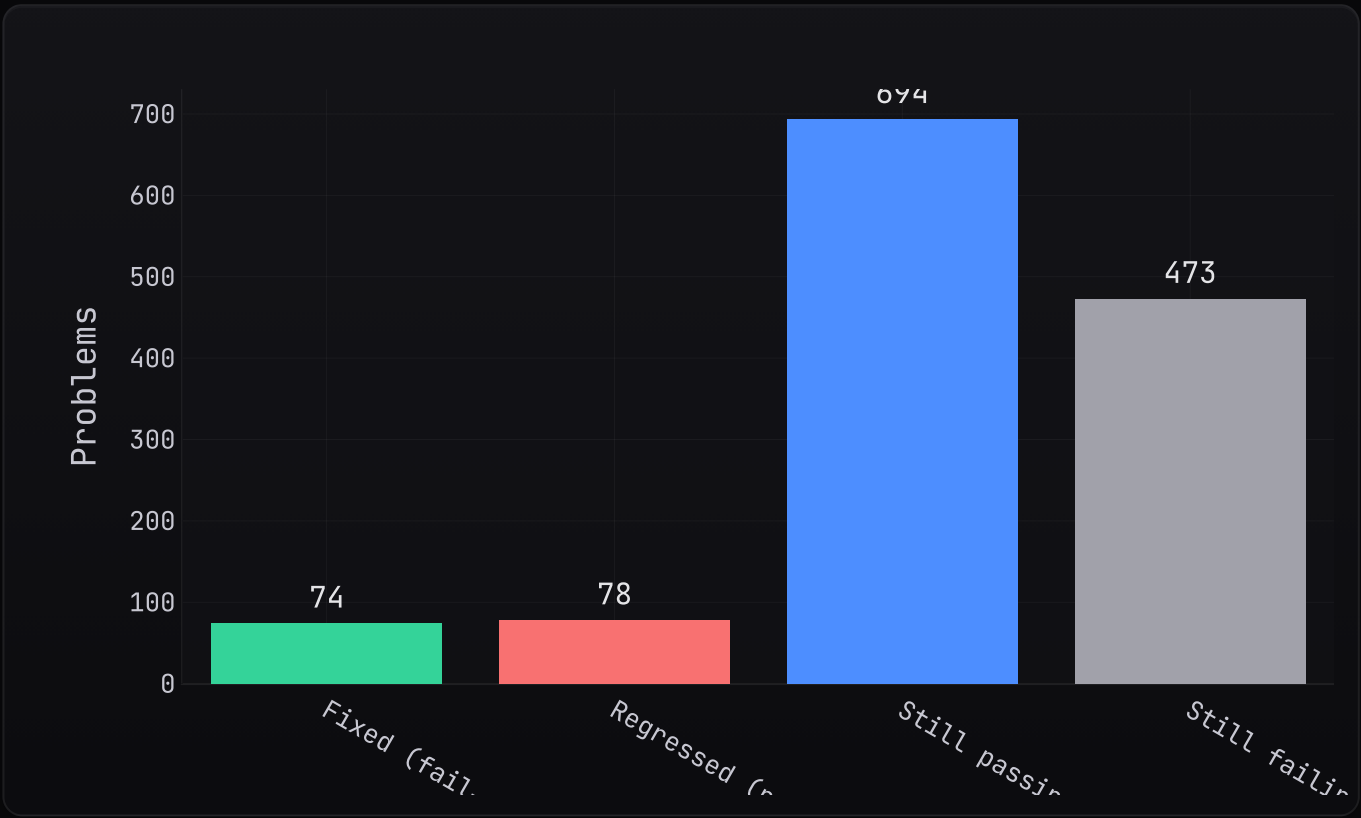


RUN 4 · GSM8K · GEMMA-2-2B-IT · UNLOCKED BUDGET

The one that learns

Same recipe as Run 3 with lr×4, β÷10, steps×2. +1.8pp — first signal outside noise.

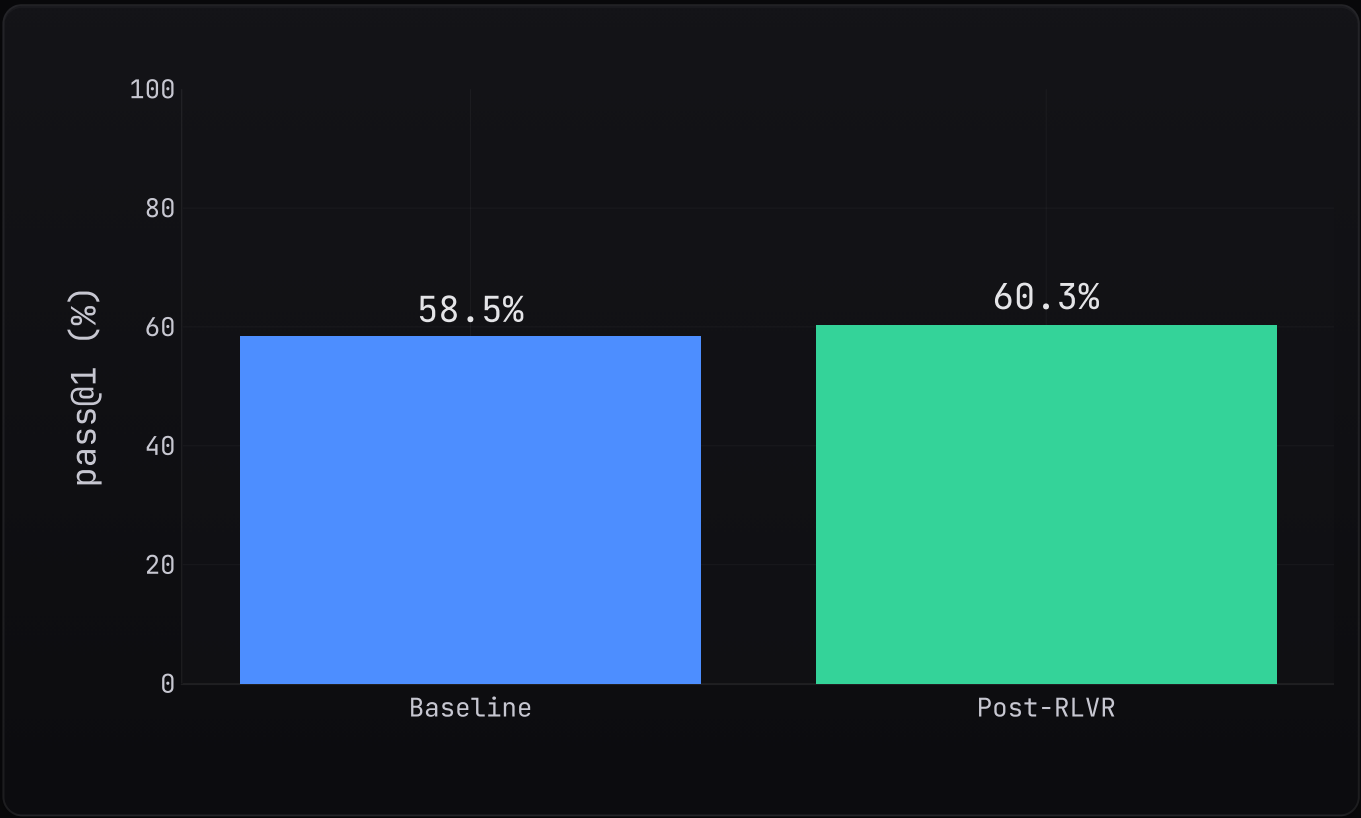
Identical model, G, and reward setup as Run 3 — the only changes are to the policy update budget: lr=2e-5 (4×), β=0.005 (10× smaller), 400 steps (2×). KL max jumped to 0.032 — 29× Run 3's ceiling. pass@1 went 58.53% → 60.35%. 120 problems fixed, 96 regressed, net +24.

58.53%	60.35%	+1.82pp
BASELINE PASS@1	POST-RLVR PASS@1	Δ
400	0.0323	
TRAINING STEPS	KL MAX	

Recipe

BASE MODEL	google/gemma-2-2b-it
GENERATIONS PER PROMPT (G)	8
TEMPERATURE	1.0
TRAINING STEPS	400
LEARNING RATE	2e-05
KL COEFFICIENT B	0.005
LORA R / A	16 / 32
MAX COMPLETION TOKENS	320
VARIANCE BAND (PREDICTED)	98.5% mixed-reward

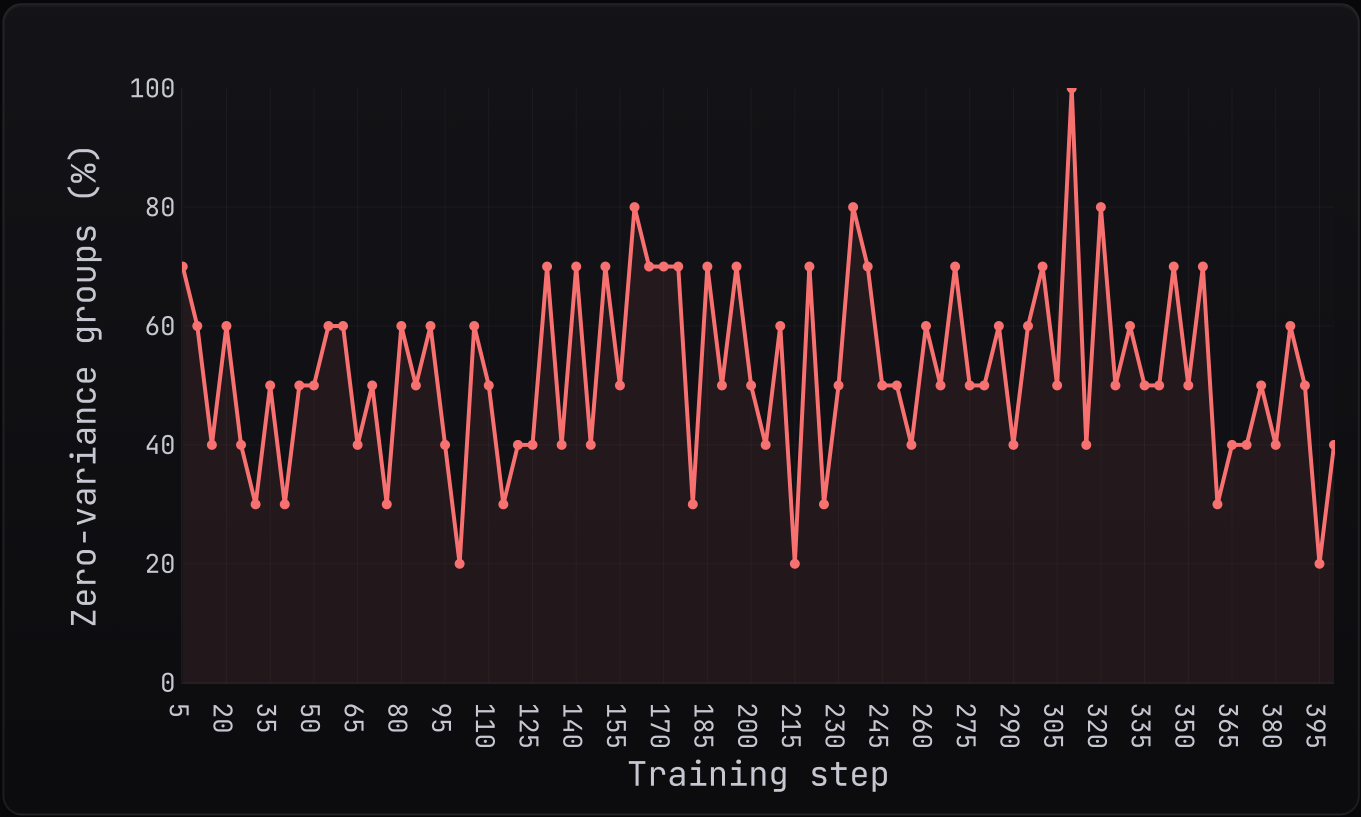
pass@1 on openai/gsm8k test (1,319)



Reward (fraction of completions with correct answer)

Reward-variance diagnostic · `frac_reward_zero_std`

Percent of GRPO groups where all G completions got the same reward. These groups have zero advantage and contribute no gradient. Run average: **52.2%**.



KL divergence from reference policy

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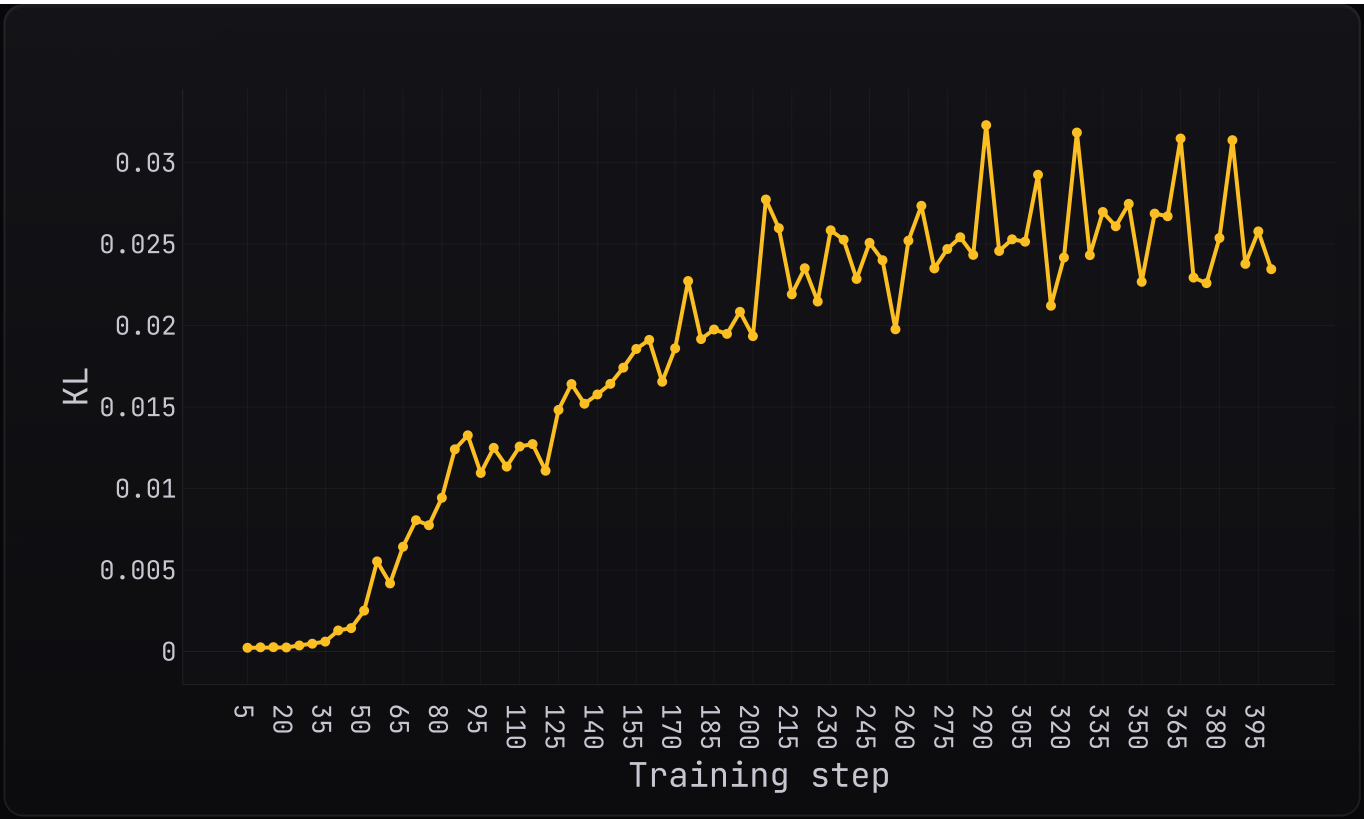

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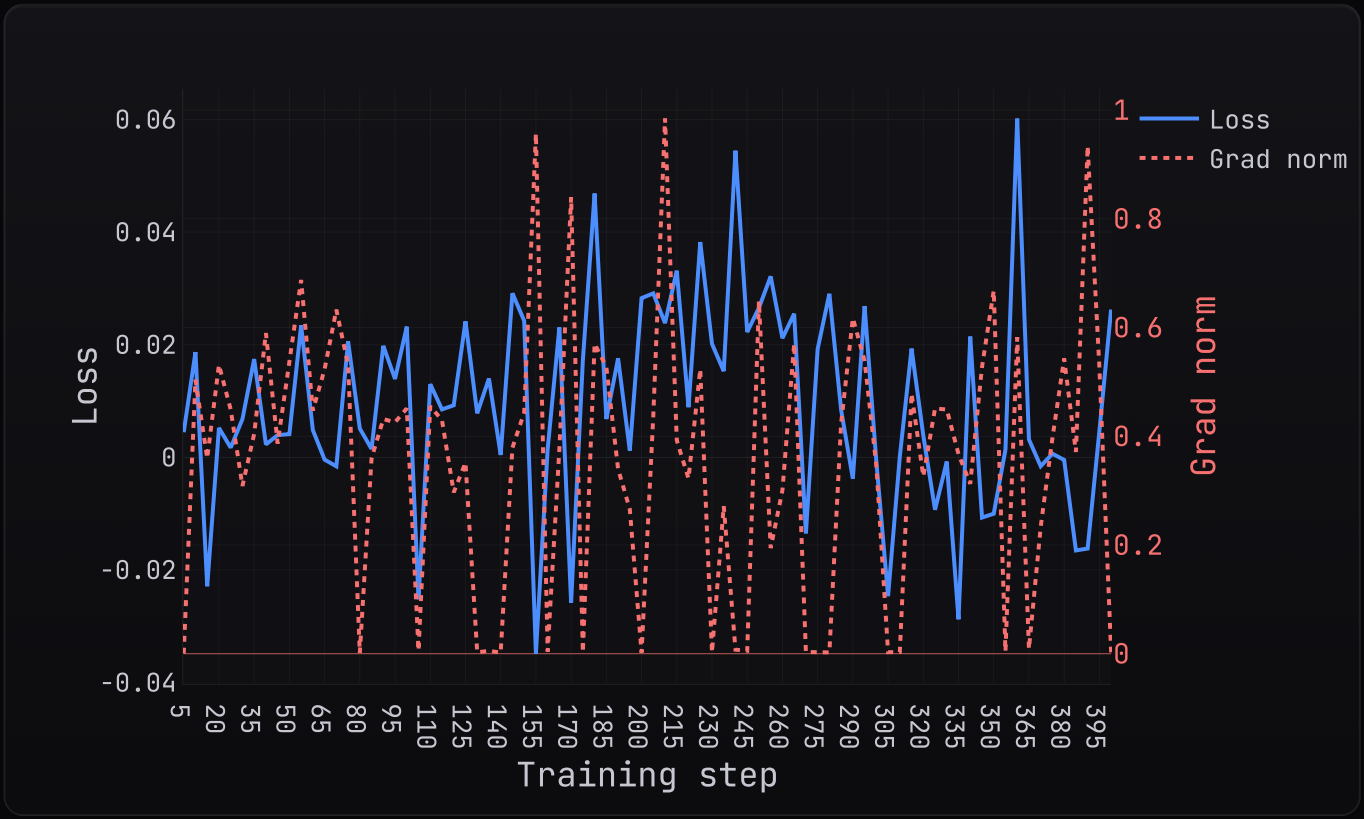

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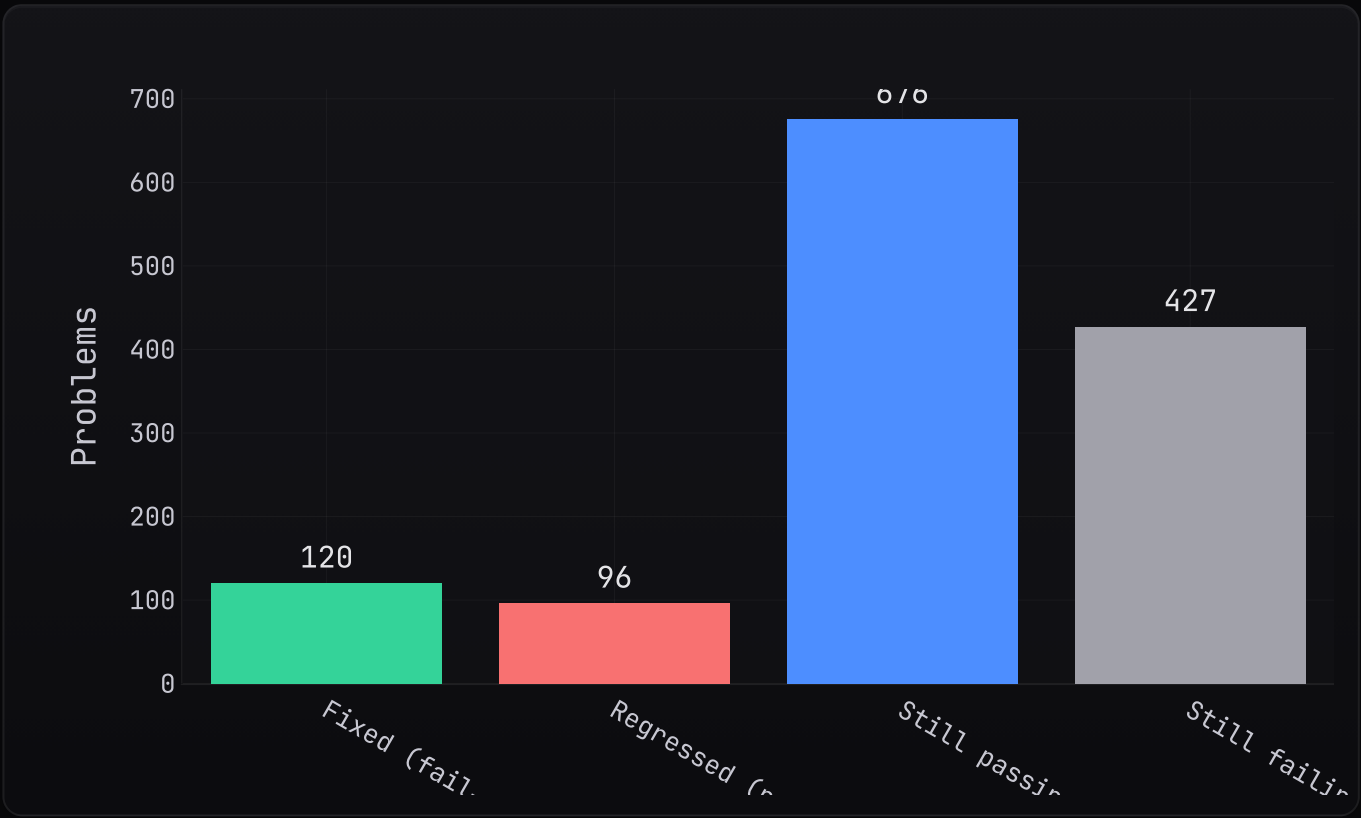

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